Johnson Jasson

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Education

Alfred University Alfred, NY
Bachelor of Science in Electrical and Electronics Engineering; Double Major in Mathematics *Aug 2025 – May 2029*

- **Honors & Grades:** Presidential Scholarship, Dean's List Fall 2025, GPA 4.00/4.00 (Fall 2025)
- **Relevant Courses:** Emerging Nanoelectronics (ELEC 453), Battery Machine Learning (ENGR 472), Object-Oriented Programming (C/C++), Multivariable Calculus (Calculus III), Differential Equations, General Chemistry, Physics, Advanced Electric Transmission (ELEC 454), Computer-Aided Programming for Engineers (MATLAB), Statics, Computer-Aided Drafting for Electrical and Electronics Engineers (CAD, SOLIDWORKS), Electrical Circuits.

CodePath
Data Structures and Algorithms | Technical Interview Prep *Sep 2025 – Dec 2025*

- **Skills:** Python, C++
- Developed proficiency in Python while studying core data structures and algorithms, including arrays, strings, linked lists, trees, recursion, and time/space complexity analysis.

Technical Skills

Programming Languages: C, C++, Python, MATLAB, Verilog
Hardware & Engineering: Digital Logic, Breadboard Prototyping, Transistors and Diodes
CAD & Engineering Tools: SolidWorks, Electrical AutoCAD, GIS (basic)
Software & Productivity: Google Colab, LaTeX, MS Office (Word, Excel, PowerPoint), MobaXterm, CRYSTAL23
Instrumentation: Oscilloscope, Digital Multimeter (DMM), Clamp Meter, Ohmmeter, Soldering Tools
Interests & Exploration: Artificial Intelligence, Machine Learning, Computer Vision, Generative AI (Prompt Engineering)

Work Experience

Electrical Engineering Intern Dar es Salaam, Tanzania
Tanzania Electric Supply Company Limited (TANESCO) *Jul 2024 – Sep 2024*

- **Skills:** Substation, AutoCAD, GIS, Planning and Survey, Troubleshooting
- Restored optimal efficiency of a 33kV transformer serving a metro area of 20 million residents, achieving measurable performance gains by conducting in-depth diagnostics and fault isolation on high-voltage distribution infrastructure.
- Optimised power distribution across service zones by designing 230V service lines in AutoCAD in full compliance with IEC standards, resulting in a streamlined and code-compliant network layout.
- Reduced Industry Packaging Limited's (IPL) electricity costs by 60% by diagnosing a critical power factor deficiency in their capacitor bank installations and implementing corrective measures to achieve a 0.95 PF target, eliminating significant reactive power losses.
- Ensured 100% regulatory compliance on project rollouts, as validated by successful field implementation sign-offs, by preparing thorough compliance documentation and collaborating directly with field engineers throughout execution.

Engineering Laboratory Support Staff Alfred, NY
Student Engineering Project Laboratory (STEP-LAB), Alfred University *Sep 2025 – Present*

- Extended vehicle range by 25+ miles and cut operational costs significantly compared to a 25 mpg gasoline baseline, by leading the Electric Vehicle Club's engine conversion project, replacing a combustion engine with a fully integrated EV drivetrain, resulting in zero direct emissions and measurable reductions in noise and air pollution by 35%.
- Designed and validated prototype components for student engineering projects with precise specifications, by utilising SolidWorks for 3D modelling and proper dimensioning, accelerating project development cycles and reducing design errors.
- Strengthened hands-on technical proficiency across electrical and mechanical systems by collaborating with engineering students and faculty on different projects, gaining direct operating experience with drilling, shearing, sharpening, and power tool machinery.
- Maintained a 100% safety compliance record across all lab sessions by enforcing established safety protocols, conducting routine equipment checks, and keeping the lab environment consistently organised and hazard-free.

Research & Projects

Effect of Temperature on Organic Semiconductors Electron Coupling Alfred, NY
Knepp Lab, Alfred University *Jan 2026 – Present*

- **Skills:** Linux, HPC (High-Performance Computing), Computational Chemistry, CRYSTAL23, QHA
- Investigates organic semiconductors by examining how temperature (T) influences electronic coupling (V) using the quasi-harmonic approximation (QHA), to understand their potential applications in the semiconductor industry.

AI-Powered Digital Twin for Smart Battery Management in Renewable Microgrids Alfred, NY
Alfred University *Oct 2025*

- Improved battery efficiency by 30% in microgrid applications by integrating self-learning Battery Management Systems (BMS) that addressed instability in renewable energy systems through real-time adaptive optimisation.
- Advanced predictive accuracy for battery State of Health (SOH) and State of Charge (SOC) by researching and developing an AI-driven digital twin model utilising Convolutional Neural Networks (CNNs), enabling more reliable battery lifecycle forecasting.
- Successfully presented research findings to faculty, industry experts, and representatives from GE-Vernova, EDIBON, NYSEG, and EPRI at the AU Energy Conference 2025, demonstrating project impact to a high-profile audience of energy sector leaders.

Wi-Fi Smart ESP32 Switch | Personal Project

- **Skills:** Arduino, Electric Circuits, C++, Breadboard Prototyping
- Enabled full remote ON/OFF device control from a mobile phone by designing and building a functional ESP32-based Wi-Fi smart switch, delivering a complete end-to-end IoT hardware-software solution.
- Achieved reliable wireless GPIO control by implementing embedded firmware in C/C++ using the Arduino framework to host an HTTP server, translating mobile network requests into real-time hardware output responses.
- Reduced development errors and ensured stable device performance by systematically debugging and validating hardware wiring, USB drivers, serial communication, and network connectivity, resulting in a fully functional and deployment-ready prototype.

Battery Life Prediction with Machine Learning Alfred, NY
Alfred University *Aug 2025 – Dec 2025*

- **Skills:** Machine Learning, Python, Google Colab, Data Analysis
- Achieved the lowest predictive error among 14 machine learning models by identifying CNN as the top-performing approach with an RMSE of 0.085, closest to the zero target, through rigorous benchmarking of approaches including Random Survival Forest (RSF), driving data-informed model selection for battery State of Health (SOH) prediction.
- Forecasted battery lifespan degradation patterns by analysing multi-cycle battery data, contributing quantifiable insights to sustainable energy storage research and advancing the reliability of next-generation battery systems.

Leadership & Involvement

Vice President of ColorStack Chapter Alfred, NY
Alfred University *Jan 2026 – Present*

- **Skills:** Leadership, Effective Decision-Making, Communication
- Strengthened club operations and leadership accountability by supporting the President in managing ColorStack's organisational structure, coordinating team leaders, and ensuring consistent delivery of member-facing initiatives across all departments.
- Elevated the professional profiles of student members by leading resume and LinkedIn optimisation workshops; guiding students in crafting polished LaTeX-formatted resumes and standout LinkedIn profiles, directly improving their recruitment readiness.
- Increased student access to internship opportunities by organising and hosting career development workshops covering internship application strategies, networking techniques, and opportunity sharing; serving both Computer Science and Engineering student communities.
- Expanded members' professional networks by facilitating intentional networking sessions and connecting students with industry contacts, recruiters, and peers to strengthen their career pipelines.

President, NSBE Chapter Alfred, NY
Alfred University *Jan 2026 – Present*

- **Skills:** Leadership, Communication, Teamwork
- Launched and led the campus chapter, increasing awareness of NSBE and creating a supportive platform for Black engineering students to showcase and develop their skills.